

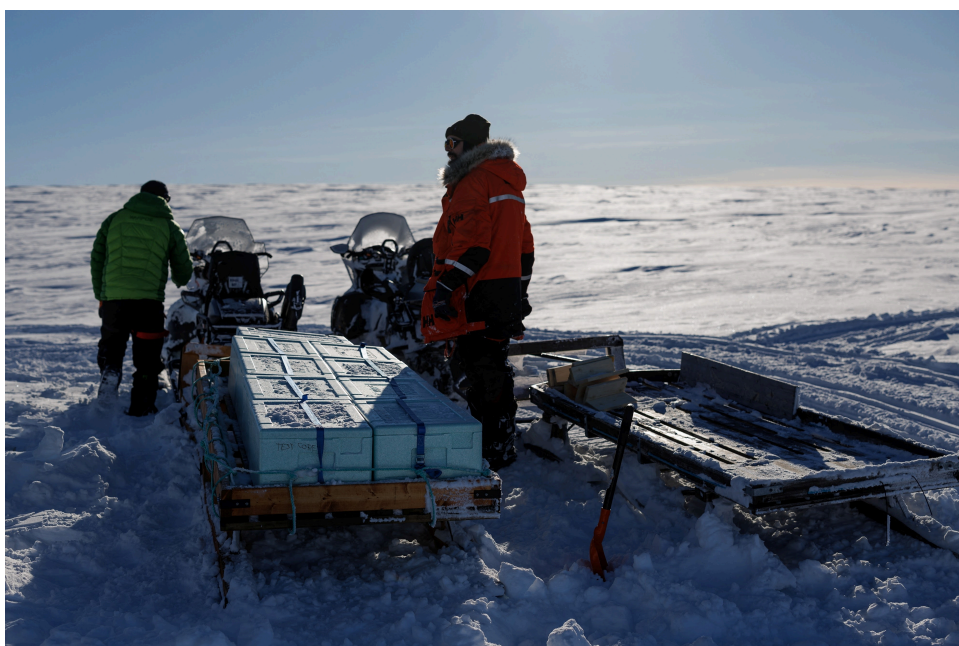
Polar Science Portal SIH Guide

<https://gemini.google.com/app/6c14386f9b96279f>

User prompt: integrated polar science outreach , knowledge repositoryb and media dissemination portal is a PS for SIH . explain me how scientists research like collect database, videos, images, graphs and all other data

Response: For a Smart India Hackathon (SIH) Problem Statement on an **Integrated Polar Science Outreach, Knowledge Repository, and Media Dissemination Portal** (typically issued by organizations like NCPOR / Ministry of Earth Sciences), understanding the actual scientist workflow is essential.

Polar research spans the Arctic, Antarctic, and the Himalayas (the Third Pole). Scientists don't just collect dry numerical spreadsheets—they gather massive amounts of raw, multimodal media and scientific datasets across extreme environments.



1. How Polar Scientists Collect Raw Data in the Field

Physical Samples & Observational Data

- **Ice Cores & Snow Samples:** Drilled hundreds of meters into glaciers. The trapped air bubbles reveal atmospheric conditions from thousands of years ago.
- **Oceanographic Data:** Deployed CTD (Conductivity, Temperature, Depth) rosettes and Autonomous Underwater Vehicles (AUVs) measure salinity, temperature, and currents under ice shelves.
- **Biological Monitoring:** Tagging marine fauna (penguins, seals, whales) and taking soil/microbial samples to track biodiversity and ecosystems.

High-Resolution Imagery & Video Media

- **Underwater Cameras & ROVs:** Capture deep-sea fauna, ice-shelf underbellies, and seabed geology.

- **Drone (UAV) Surveying:** High-definition aerial mapping to track glacier retreat, ice-crack propagation, and wildlife colony sizes.
- **GoPros & Expedition Footage:** Scientists record daily fieldwork, weather challenges, instrument deployments, and station life (e.g., at India's *Bharati*, *Maitri*, or *Himadri* stations).

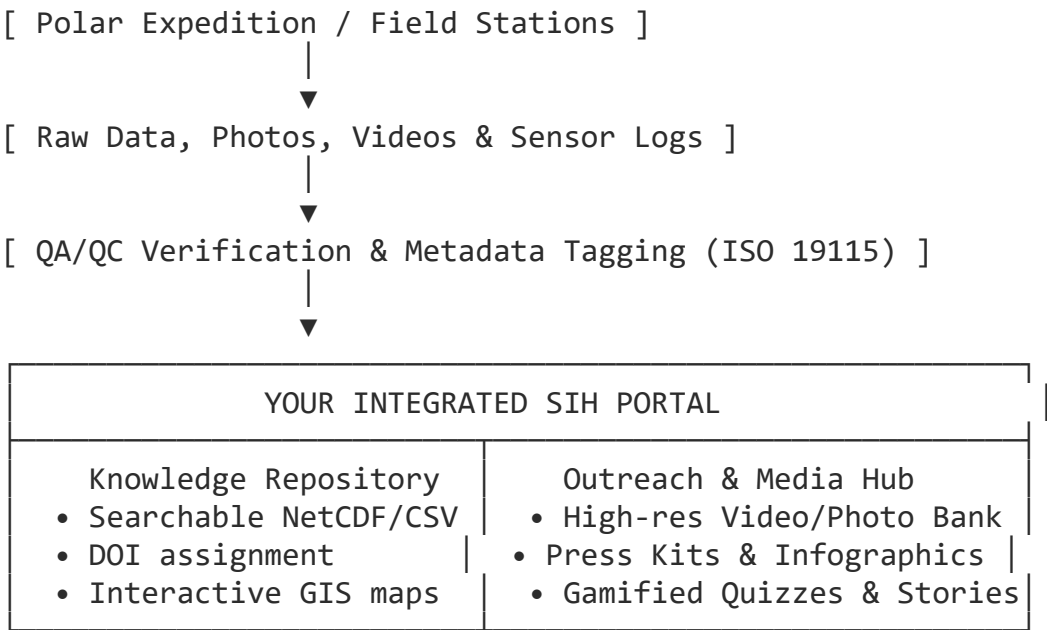
Remote Sensing & Automated Sensor Networks

- **Satellite Telemetry:** SAR (Synthetic Aperture Radar) and optical imagery tracking sea-ice extent and ice sheet thickness.
- **Automatic Weather Stations (AWS):** Transmit continuous real-time telemetry (wind speed, solar radiation, temperature) via satellite links.

2. Data Types You Must Handle in Your Repository

Data Category	Specific Formats / Inputs	Use Case in Your Portal
Numerical & Tabular	CSV, NetCDF, HDF5, ASCII, GeoTIFF	Downloadable for researchers; interactive plot generators for students.
Multimedia Assets	RAW/JPEG photos, 4K video clips, drone footage	Media dissemination, press releases, public galleries, virtual tours.
Spatial & GIS Data	Shapefiles (.shp), GeoJSON, KML	Interactive maps showing station locations, expedition routes, ice boundaries.
Scientific Visuals	Time-series graphs, bathymetric maps, temperature profiles	Embedded in outreach articles and educational modules.

3. The Lifecycle: From Expedition to Your SIH Portal



4. Key Architectural Features Needed for the SIH Solution

To make your portal succeed for SIH evaluation, bridge the gap between **heavy scientific research** and **engaging public outreach**:

- 1. **Metadata Standardization (ISO 19115 / FGDC):** Scientists require metadata tags (latitude, longitude, date/time, instrument ID, principal investigator). Implement auto-extraction of EXIF data from photographs and spatial bounding boxes from NetCDF files.
- 2. **Dual-Layer Access Control:**
 - *Public/Outreach Layer:* Fast-loading web-optimized images, streamable 1080p videos, simple interactive charts, and press release kits.
 - *Researcher/Repository Layer:* Raw dataset downloads (NetCDF, HDF5), GIS layers, and citation metadata (DOIs).
- 3. **Interactive Data Visualization:** Build lightweight web viewers (using libraries like Chart.js, D3.js, or Leaflet/Mapbox) so users can plot temperature anomalies or view glacier retreat directly in the browser without downloading heavy software.
- 4. **Media Asset Management (MAM):** Filter media by tags (e.g., #Antarctica, #BharatiStation, #IceCore, #Wildlife). Include clear usage licensing (e.g., Creative Commons) for media outlets and educators.

User prompt: on software basis how the scientists research upon it

Response: To build an effective **Integrated Polar Science Knowledge Repository and Media Dissemination Portal** for your Smart India Hackathon (SIH) project, you need to understand how scientists handle software throughout their research.

Scientists rely on specialized open-source software, custom code, and geospatial platforms rather than standard off-the-shelf software. The software-driven workflow transforms raw expedition files into validated scientific data and media ready for your portal.

1. The Core Scientific Software Stack

Category	Primary Software & Libraries	What Scientists Use It For
Multidimensional Data	xarray, nctoolkit (Python), MATLAB, CDO/NCO	Processing 4D climate/ocean data (Latitude, Longitude, Depth, Time) stored in NetCDF (.nc) or HDF5 formats.
Geospatial & Remote Sensing	QGIS, ArcGIS Pro, GDAL, Google Earth Engine	Plotting satellite radar (SAR) images, tracking glacier retreat, and layering Shapefiles or GeoTIFFs.
3D Photogrammetry & Media	Agisoft Metashape, Pix4D, OpenCV	Stitching drone/UAV images into 3D models of ice sheets or terrain, and processing high-res field videos.

Category	Primary Software & Libraries	What Scientists Use It For
Sensor Telemetry & Ingestion	ROS (Robot Operating System), Python (pandas, numpy)	Ingesting real-time streaming data from Automatic Weather Stations (AWS) or underwater ROVs.

2. Step-by-Step Software Workflow of Polar Researchers

Step 1: Ingestion & Raw File Parsing

Scientists return from missions with mixed files (raw drone footage, temperature logs, NetCDF files, geotagged photos).

- **Data Processing:** They use **Python (xarray)** or **MATLAB** to parse multi-gigabyte binary files, slice coordinates, and smooth out sensor noise.
- **Media Processing:** Drones output hundreds of overlapping images. Photogrammetry software converts these images into 3D point clouds and high-resolution digital elevation maps (DEMs).

Step 2: Spatial & Temporal Standardization

Because Earth isn't flat—and projection distorts significantly at the North and South Poles—scientists must reproject all geospatial data into Polar Stereographic projections using tools like **QGIS** or **GDAL**.

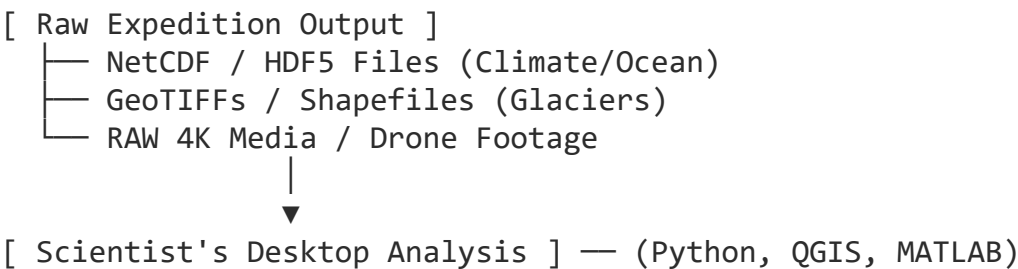
Step 3: Automated Metadata Extraction

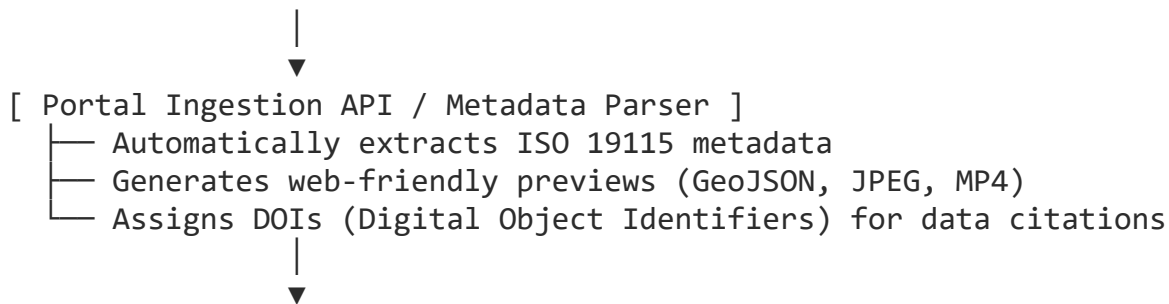
Before uploading data to a central portal, researchers write scripts to attach standard metadata (such as ISO 19115 standards). Python scripts scan file headers to automatically extract:

- **Spatial Coordinates:** Bounding box (Latitude/Longitude boundaries).
- **Temporal Boundaries:** Start time and end time of the expedition.
- **EXIF Data:** Camera orientation, camera model, and GPS tags from field media.

3. How Your SIH Portal Must Integrate with This Software Workflow

Your SIH solution needs to act as a seamless bridge between heavy-duty scientific software and public web browsers:





YOUR SIH PORTAL	
Knowledge Repository • Search NetCDF/CSV data • In-browser map plotters • API access for scripts	Outreach & Media Portal • Stream 1080p/4K Video • Download Press Kits • High-res Photo Libraries

Key Technical Features to Build in Your Software Architecture

1. **NetCDF / GeoJSON Web Viewer:** Integrate JavaScript plotting libraries (like `Plotly.js`, `Leaflet`, or `Mapbox GL`) so visitors can interact with 3D glacier maps or temperature graphs directly on the site without needing QGIS or MATLAB installed on their computers.
2. **Metadata Auto-parsing Backend:** Write a Python (`FastAPI` or `Django`) backend that runs `netcdf4` and `EXIFRead` libraries. When a scientist drops a file into your repository upload form, your portal should automatically fill in spatial, temporal, and sensor tags.
3. **Media Transcoding Pipeline:** Heavy 4K GoPro and drone videos must automatically convert to compressed `.mp4` / `.webm` streams for public outreach while storing the uncompressed original in cloud storage (e.g., `AWS S3` or `MinIO`) for research downloads.

Sources

1. [nctoolkit: A Python package for netCDF analysis and post-processing](#)
2. [Data Tools | National Snow and Ice Data Center](#)